Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software engineer with 2+ years building **performance-critical systems**, **data pipelines**, and **ML infrastructure**, with strong foundations in algorithms, systems optimization, and cross-functional collaboration across architecture and engineering teams.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Computer Architecture, Object-Oriented Programming

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **performance-optimized platform services** for 500+ users, reducing request latency from **1.2s to 350ms (71%)** by profiling execution paths, optimizing query patterns, and tuning memory and I/O access for analytics workloads.
- Implemented **automated validation and pipeline checks** for 5,000+ monthly jobs, reducing downstream failures through earlier detection of regressions, enabling faster root-cause analysis and debug cycles.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **concurrent batch-processing engine** handling 10,000+ daily requests, optimizing throughput through multithreading, task scheduling, and resource-efficient execution paths in production workloads.
- Built **multimodal ML workflows** integrating vision models and OCR, profiling inference latency and reducing feedback turnaround by **60%** while maintaining consistent performance under concurrent load.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **large-scale experimentation pipelines** across 100,000+ trajectory records for autonomous driving research, enabling reproducible algorithm evaluation and high-confidence benchmarking with PyTorch and model training infrastructure.
- Designed **spatial-temporal encoder architecture** integrating HD-map geometry with trajectory context via multi-head attention, improving closed-loop planning stability by **15%** in simulation evaluation.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **API and ETL microservices** using AWS infrastructure, improving response time by **30%** through algorithmic optimization, caching strategies, and profiling-driven performance tuning.
- Maintained **20+ automated test suites** using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%** and enabling faster correlation and debug cycles across cross-platform validation workflows.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **real-time synchronization modules** for 2M+ records using WebSockets, optimizing throughput consistency and reducing sync conflicts by **33%** through efficient state management and memory handling.
- Automated **Docker-based build and test infrastructure** with Jenkins, cutting release time by **50%** and improving deployment consistency across multi-environment validation pipelines.

Skills

Languages: **C++**, **Python**, Java, SQL, Bash, JavaScript/TypeScript
Systems & Infrastructure: **Linux**, Docker, Kubernetes, Git, CI/CD, Jenkins, AWS
Performance: Profiling, benchmarking, latency optimization, concurrent programming, memory tuning
ML Infrastructure: PyTorch, Transformers, model training pipelines, experiment tracking
Databases: PostgreSQL, MySQL, query optimization, indexing, relational schema design

Projects

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, OS Internals*

- Built a **scheduling algorithm simulator** implementing FCFS, SJF, SRT, RR, and MLQ, reducing average turnaround time by **25%** across 1K+ simulated process combinations with Gantt-chart visualization and performance metrics.

Chat2SQL Code | *Java, TypeScript, SQL, Docker, Maven*

- Built an **AI-powered SQL workspace** supporting 10+ database systems, working across Java backend and TypeScript frontend with Docker-based development workflows and cross-platform compatibility.

Linguistix Code | *Python, AWS, Docker, CI/CD*

- Built a **containerized REST processing API** on AWS CodePipeline and ECS/Fargate with automated validation gates, improving reliability and streamlining troubleshooting across production deployment workflows.